

✓ SoC-25 OpenROAD PU17 uart demo.

(C) DJ Greaves, University of Cambridge Computer Laboratory, Feb 2025.

This demo can be run on your own computer, or here in Google Colab.

Acknowledgements

- The set-up code in the first block here is copied from widely-shared [Chips Alliance's](#) using the Apache 2.0 license.

```
#
# SoC-25 OpenROAD PU17 uart demo
#
!date
import os
import pathlib

# Firstly - install the tools we need in the Colab VM.
#
# NB: The core of this set-up code was copied from the Chips Alliance Silicon Notebooks
# under the Apache 2.0 license.

!sudo apt-get update
!sudo apt-get install time

!curl -Ls https://micro.mamba.pm/api/micromamba/linux-64/latest | tar -xj bin/micromamba
conda_prefix_path = pathlib.Path('conda-env')
CONDA_PREFIX = str(conda_prefix_path.resolve())
bin/micromamba create --yes --prefix $CONDA_PREFIX
!echo 'python ==3.7*' >> {CONDA_PREFIX}/conda-meta/pinned
!CI=0 bin/micromamba install --yes --prefix $CONDA_PREFIX \
    --channel litex-hub \
    --channel main \
    openroad \
    yosys \
    klayout

#
!sudo pip install gdstk
#
!pip3 install pyyaml
PATH = os.environ['PATH']
%env CONDA_PREFIX={CONDA_PREFIX}
%env PATH={CONDA_PREFIX}/bin:{PATH}
```

 Sun Apr 27 09:21:12 AM UTC 2025

```
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:3 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease [3,632 B]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:5 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Get:6 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64 InRelease [1,581 B]
Hit:7 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
Get:8 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:9 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy InRelease [24.3 kB]
Hit:10 https://ppa.launchpadcontent.net/ubuntuugis/ppa/ubuntu jammy InRelease
Get:11 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [8,868 kB]
Get:12 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64 Packages [1,607 kB]
Get:13 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [2,704 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,543 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,154 kB]
Get:16 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy/main amd64 Packages [47.4 kB]
Fetched 18.3 MB in 9s (2,007 kB/s)
Reading package lists... Done
W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (sources.list e
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  time
0 upgraded, 1 newly installed, 0 to remove and 35 not upgraded.
Need to get 46.7 kB of archives.
After this operation, 129 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 time amd64 1.9-0.1build2 [46.7 kB]
Fetched 46.7 kB in 0s (123 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78, <> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package time.
(Reading database ... 126333 files and directories currently installed.)
Preparing to unpack .../time_1.9-0.1build2_amd64.deb ...
Unpacking time (1.9-0.1build2) ...
Setting up time (1.9-0.1build2) ...
Processing triggers for man-db (2.10.2-1) ...
Empty environment created at prefix: /content/conda-env

Pinned packages:

- python==3.7*

Transaction

Prefix: /content/conda-env

Updating specs:

- openroad
- yosys

!date
# Check tools are present and on the execution path.
!openroad -version
!yosys --version
!klayout -zz -v
# Fetch the workshop starter pack and unzip it.
# Note, this refers to a temporary web page that worked for me. You can instead fetch
# it from the edX materials and then reupload it to Google Colab if you are using Colab.
!wget https://www.cl.cam.ac.uk/users/djg11/soc25-starter-packs/soc25-skills4-ws-starterpack-openroad.zip
# Unzip workshop starter pack
!unzip soc25-skills4-ws-starterpack-openroad.zip
```

```

Sun Apr 27 09:33:18 AM UTC 2025
2.0-12381-g01bba3695
Yosys 0.38+92 (git sha1 84116c9a3, x86_64-conda-linux-gnu-cc 11.2.0 -fvisibility-inlines-hidden -fmessage-length=0 -march=nocona -mtune=haswell -ftree-vectorize -fPIC -fstac
Layout 0.28.17
--2025-04-27 09:33:19-- https://www.cl.cam.ac.uk/users/djg11/soc25-starter-packs/soc25-skills4-ws-starterpack-openroad.zip
Resolving www.cl.cam.ac.uk (www.cl.cam.ac.uk)... 128.232.0.20, 2a05:b400:110::80:14
Connecting to www.cl.cam.ac.uk (www.cl.cam.ac.uk)|128.232.0.20|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.cl.cam.ac.uk/~djg11/soc25-starter-packs/soc25-skills4-ws-starterpack-openroad.zip [following]
--2025-04-27 09:33:19-- https://www.cl.cam.ac.uk/~djg11/soc25-starter-packs/soc25-skills4-ws-starterpack-openroad.zip
Reusing existing connection to www.cl.cam.ac.uk:443.
HTTP request sent, awaiting response... 200 OK
Length: 4451033 (4.2M) [application/zip]
Saving to: 'soc25-skills4-ws-starterpack-openroad.zip'

soc25-skills4-ws-st 100%[=====] 4.24M 5.74MB/s in 0.7s

2025-04-27 09:33:20 (5.74 MB/s) - 'soc25-skills4-ws-starterpack-openroad.zip' saved [4451033/4451033]

Archive: soc25-skills4-ws-starterpack-openroad.zip
creating: soc25-skills4-ws-starterpack-openroad/
creating: soc25-skills4-ws-starterpack-openroad/uart16-demo/
inflating: soc25-skills4-ws-starterpack-openroad/uart16-demo/config-soc25-uart-openroad-demo.mk
inflating: soc25-skills4-ws-starterpack-openroad/uart16-demo/constraint-soc25-uart-openroad-demo.sdc
inflating: soc25-skills4-ws-starterpack-openroad/uart16-demo/uart16.v
creating: soc25-skills4-ws-starterpack-openroad/pictures/
inflating: soc25-skills4-ws-starterpack-openroad/pictures/openroad-flow-diag.pdf
inflating: soc25-skills4-ws-starterpack-openroad/pictures/ColabNoteBook-screenshot.png
inflating: soc25-skills4-ws-starterpack-openroad/pictures/openroad-flow-diag.png
inflating: soc25-skills4-ws-starterpack-openroad/pictures/openroad-gomactech19.pdf
inflating: soc25-skills4-ws-starterpack-openroad/pictures/OpenRoad-gomactech-image.png
inflating: soc25-skills4-ws-starterpack-openroad/pictures/uart16-layout-shot.png
inflating: soc25-skills4-ws-starterpack-openroad/pictures/uart16-openROAD-logfile-output.pdf
inflating: soc25-skills4-ws-starterpack-openroad/pictures/uart-closeup-obj.png
inflating: soc25-skills4-ws-starterpack-openroad/copyright_message.txt
inflating: soc25-skills4-ws-starterpack-openroad/openroad-flow-diag.pdf

# Get the OpenROAD flow scripts
!git clone https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts.git

Cloning into 'OpenROAD-flow-scripts'...
remote: Enumerating objects: 59255, done.
remote: Counting objects: 100% (8031/8031), done.
remote: Compressing objects: 100% (611/611), done.
remote: Total 59255 (delta 7621), reused 7420 (delta 7420), pack-reused 51224 (from 4)
Receiving objects: 100% (59255/59255), 847.54 MiB | 25.15 MiB/s, done.
Resolving deltas: 100% (43659/43659), done.
Updating files: 100% (2445/2445), done.

# Before using Google Colab, I had run the OpenROAD yosys stage on my own workstation
# and then transferred my intermediate results to the cloud using the following zip.
# It is entirely up to you what balance between local and cloud mix you use. The openROAD
# makefile should automatically join the gaps based on file time stamps.
!date; pwd
!cd results/nangate45; wget https://www.cl.cam.ac.uk/users/djg11/soc25-starter-packs/UART16-yosys-output.zip
!cd results/nangate45; unzip -o UART16-yosys-output.zip
!cd results/nangate45; touch UART16/base/* # Bring it up-to-date
!cd results/nangate45; ls -l UART16/base/

Sun Apr 27 10:35:34 AM UTC 2025
/content/OpenROAD-flow-scripts/flow
--2025-04-27 10:35:34-- https://www.cl.cam.ac.uk/users/djg11/soc25-starter-packs/UART16-yosys-output.zip
Resolving www.cl.cam.ac.uk (www.cl.cam.ac.uk)... 128.232.0.20, 2a05:b400:110::80:14
Connecting to www.cl.cam.ac.uk (www.cl.cam.ac.uk)|128.232.0.20|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.cl.cam.ac.uk/~djg11/soc25-starter-packs/UART16-yosys-output.zip [following]
--2025-04-27 10:35:35-- https://www.cl.cam.ac.uk/~djg11/soc25-starter-packs/UART16-yosys-output.zip
Reusing existing connection to www.cl.cam.ac.uk:443.
HTTP request sent, awaiting response... 200 OK
Length: 38615 (38K) [application/zip]
Saving to: 'UART16-yosys-output.zip.1'

UART16-yosys-output 100%[=====] 37.71K --.-KB/s in 0.09s

2025-04-27 10:35:35 (410 KB/s) - 'UART16-yosys-output.zip.1' saved [38615/38615]

Archive: UART16-yosys-output.zip
inflating: UART16/base/1_synth.sdc
inflating: UART16/base/1_yosys.v
inflating: UART16/base/1_synth.v
extracting: UART16/base/clock_period.txt
inflating: UART16/base/mem.json
inflating: UART16/base/1_synth.rtlil
total 512
-rw-r--r-- 1 root root 57358 Apr 27 10:35 1_1_yosys.v
-rw-r--r-- 1 root root 303856 Apr 27 10:35 1_synth.rtlil
-rw-r--r-- 1 root root 456 Apr 27 10:35 1_synth.sdc
-rw-r--r-- 1 root root 57358 Apr 27 10:35 1_synth.v
-rw-r--r-- 1 root root 5 Apr 27 10:35 clock_period.txt
-rw-r--r-- 1 root root 82679 Apr 27 10:35 mem.json

!date ; pwd
# Set the main environment variables
%env YOSYS_EXE=yosys
%env OPENROAD_EXE=openroad
%env KLAYOUT_EXE=klayout
# Set the design-specific environment variables
%env SOC25_STARTPACK=/content/soc25-skills4-ws-starterpack-openroad
%env DESIGN_HOME=$(SOC25_STARTPACK)/uart16-demo
%env DESIGN_CONFIG=$(DESIGN_HOME)/config-soc25-uart-openroad-demo.mk
%env PLATFORM=nangate45
%env DESIGN_NAME=UART16

# Needed to install again for some reason?
# !pip3 install pyyaml

# And finally launch the OpenROAD flow Makefile.
%env SYNTH_OPERATIONS_ARGS=""
!cd /content/OpenROAD-flow-scripts/flow
!make do-floorplan do-place do-cts do-route do-finish
!echo Done OpenROAD Makefile for SoC-25

Sun Apr 27 10:43:25 AM UTC 2025
/content/OpenROAD-flow-scripts/flow
env: YOSYS_EXE=yosys
env: OPENROAD_EXE=openroad
env: KLAYOUT_EXE=klayout
env: SOC25_STARTPACK=/content/soc25-skills4-ws-starterpack-openroad
env: DESIGN_HOME=$(SOC25_STARTPACK)/uart16-demo

```

```
env: DESIGN_CONFIG=$(DESIGN_HOME)/config-soc25-uart-openroad-demo.mk
env: PLATFORM=nangate45
env: DESIGN_NAME=UART16
env: SYNTH_OPERATIONS_ARGS=""
/content/OpenROAD-flow-scripts/flow
/content/OpenROAD-flow-scripts/flow/scripts/flow.sh 2_1_floorplan floorplan
Running floorplan.tcl, stage 2_1_floorplan
[INFO ORD-0030] Using 2 thread(s).
[INFO ODB-0222] Reading LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.tech.lef
[INFO ODB-0223] Created 22 technology layers
[INFO ODB-0224] Created 27 technology vias
[INFO ODB-0226] Finished LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.tech.lef
[INFO ODB-0222] Reading LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.macro.mod.lef
[INFO ODB-0225] Created 135 library cells
[INFO ODB-0226] Finished LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.macro.mod.lef
[WARNING STA-0347] current_design for other than top cell not supported.
source /content/OpenROAD-flow-scripts/flow/platforms/nangate45/setRC.tcl

=====
Floorplan check_setup
-----
Warning: There are 41 unlocked register/latch pins.
Warning: There are 50 unconstrained endpoints.
number instances in verilog is 287
[WARNING IFP-0028] Core area lower left (1.000, 1.000) snapped to (1.140, 1.400).
[INFO IFP-0001] Added 25 rows of 186 site FreePDK45_38x28_10R_NP_162NW_340.
source /content/OpenROAD-flow-scripts/flow/platforms/nangate45/make_tracks.tcl
repair_timing -verbose -setup_margin 0 -repair_tns 100
[WARNING RSZ-0021] no estimated parasitics. Using wire load models.
[INFO RSZ-0033] No hold violations found.
Default units for flow
time 1ns
capacitance 1fF
resistance 1kohm
voltage 1v
current 1mA
power 1nW
distance 1um
Report metrics stage 2, floorplan final...

=====
floorplan final report_design_area
-----
Design area 509 u^2 41% utilization.
Elapsed time: 0:00.62[h]:min:sec. CPU time: user 0.58 sys 0.05 (101%). Peak memory: 88208KB.
/content/OpenROAD-flow-scripts/flow/scripts/flow.sh 2_2_floorplan_macro macro_place
Running macro_place.tcl, stage 2_2_floorplan_macro
[INFO ORD-0030] Using 2 thread(s).
source /content/OpenROAD-flow-scripts/flow/platforms/nangate45/setRC.tcl
No macros found: Skipping macro_placement
Elapsed time: 0:00.52[h]:min:sec. CPU time: user 0.46 sys 0.05 (99%). Peak memory: 85296KB.

# Now that we have a GDS-II layout file, lets download it to our workstation or PC
# for interactive examination. So zip up all the output results for download.
!pwd
!cd ..; zip -r dump-for-download-soc25-OR-uart-demo.zip flow/results flow/UART16 flow/scripts flow/Makefile

/content/OpenROAD-flow-scripts/flow
adding: flow/results/ (stored 0%)
adding: flow/results/nangate45/ (stored 0%)
adding: flow/results/nangate45/UART16-yosys-output.zip.1 (deflated 19%)
adding: flow/results/nangate45/UART16/ (stored 0%)
adding: flow/results/nangate45/UART16/base/ (stored 0%)
adding: flow/results/nangate45/UART16/base/3_2_place_iop.tcl (deflated 88%)
adding: flow/results/nangate45/UART16/base/2_floorplan.odb (deflated 81%)
adding: flow/results/nangate45/UART16/base/clock_period.txt (stored 0%)
adding: flow/results/nangate45/UART16/base/1_1_yosys.v (deflated 89%)
adding: flow/results/nangate45/UART16/base/1_synth.v (deflated 89%)
adding: flow/results/nangate45/UART16/base/2_1_floorplan.sdc (deflated 93%)
adding: flow/results/nangate45/UART16/base/3_5_place_dp.odb (deflated 82%)
adding: flow/results/nangate45/UART16/base/updated_clks.sdc (deflated 93%)
adding: flow/results/nangate45/UART16/base/mem.json (deflated 93%)
adding: flow/results/nangate45/UART16/base/2_floorplan.sdc (deflated 93%)
adding: flow/results/nangate45/UART16/base/3_place.odb (deflated 82%)
adding: flow/results/nangate45/UART16/base/1_synth.sdc (deflated 57%)
adding: flow/results/nangate45/UART16/base/2_1_floorplan.odb (deflated 81%)
adding: flow/results/nangate45/UART16/base/4_cts.odb (deflated 82%)
adding: flow/results/nangate45/UART16/base/3_place.sdc (deflated 93%)
adding: flow/results/nangate45/UART16/base/3_4_place_resized.odb (deflated 81%)
adding: flow/results/nangate45/UART16/base/3_3_place_gp.odb (deflated 82%)
adding: flow/results/nangate45/UART16/base/2_2_floorplan_macro.tcl (stored 0%)
adding: flow/results/nangate45/UART16/base/3_2_place_iop.odb (deflated 80%)
adding: flow/results/nangate45/UART16/base/4_1_cts.odb (deflated 82%)
adding: flow/results/nangate45/UART16/base/2_2_floorplan_macro.odb (deflated 81%)
adding: flow/results/nangate45/UART16/base/1_synth.rtlil (deflated 94%)
adding: flow/results/nangate45/UART16/base/3_1_place_gp_skip_io.odb (deflated 80%)
adding: flow/results/nangate45/UART16/base/2_3_floorplan_tapcell.odb (deflated 81%)
adding: flow/results/nangate45/UART16/base/2_4_floorplan_pdn.odb (deflated 81%)
adding: flow/results/nangate45/UART16/base/4_cts.sdc (deflated 92%)
adding: flow/results/nangate45/UART16-yosys-output.zip (stored 0%)
adding: flow/UART16/ (stored 0%)
adding: flow/UART16/base/ (stored 0%)
adding: flow/UART16/base/clock_period.txt (stored 0%)
adding: flow/UART16/base/1_1_yosys.v (deflated 89%)
adding: flow/UART16/base/1_synth.v (deflated 89%)
adding: flow/UART16/base/mem.json (deflated 93%)
adding: flow/UART16/base/1_synth.sdc (deflated 57%)
adding: flow/UART16/base/1_synth.rtlil (deflated 94%)
adding: flow/scripts/ (stored 0%)
adding: flow/scripts/yosys_keep.tcl (deflated 18%)
adding: flow/scripts/macro_place_util.tcl (deflated 71%)
adding: flow/scripts/macro_place.tcl (deflated 46%)
adding: flow/scripts/add_routing_blk.tcl (deflated 63%)
adding: flow/scripts/write_ref_sdc.tcl (deflated 57%)
adding: flow/scripts/yosys_load.tcl (deflated 14%)
adding: flow/scripts/cdl.tcl (deflated 24%)
adding: flow/scripts/placement_blockages.tcl (deflated 57%)
adding: flow/scripts/abc_speed_gia_only.script (deflated 78%)
adding: flow/scripts/detail_place.tcl (deflated 47%)
adding: flow/scripts/io_placement.tcl (deflated 53%)
adding: flow/scripts/sta_synth.tcl (deflated 11%)
adding: flow/scripts/synth_metrics.tcl (deflated 16%)
adding: flow/scripts/variables.yaml (deflated 69%)
adding: flow/scripts/synth.sh (deflated 27%)
adding: flow/scripts/util.tcl (deflated 63%)

!date
import pathlib
import gdstk
import IPython.display

def gen_svg(): # This display code also based on Chips Alliance version, but not being used today anyway.
    gdss = sorted(pathlib.Path('.').glob('._results/nangate45/UART16/base/*.*gds'))
    library = gdstk.read_gds(gdss[-1])
    top_cells = library.top_level()
```

```
# this is to hide all layer labels
label_style = {(i,0): {"fill": "none", "stroke": "red", "font-size": "0px"} for i in range(256)}
top_cells[0].write_svg('UART16.svg',
                      label_style=label_style)
IPython.display.SVG('UART16.svg')

print(f'DJG end of flow')

Sun Apr 27 10:47:20 AM UTC 2025
DJG end of flow
```